David Beckwitt, Ph.D. Candidate

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Physics PhD candidate (July 2026) focused on materials structure analysis, structural modelling, and scientific ML for thin-film analysis. My work combines X-ray diffraction and reflectivity instrumentation with simulation and AI-assisted scientific-computing workflows to study materials structure and disorder.

Education

July 2026 — **University of Missouri, Columbia, MO — Ph.D. in Physics**
Dissertation: *Investigating Disorder in van der Waals Thin Films*
Advisor: Dr. Paul Miceli

May 2022 — **University of Missouri, Columbia, MO — M.S. in Physics**

May 2020 — **Missouri State University, Springfield, MO — B.S. in Physics**
Minor: Mathematics, Chemistry

Ongoing Role

2021–Present — **Graduate Research Assistant**, University of Missouri
Focus: X-ray diffraction instrumentation, GIWAXS simulation, thin-film growth, and AI-assisted scientific-computing workflows.

- **Lab operations:** Operate and maintain custom-built in-house x-ray diffraction and reflectivity systems for materials characterization.
- **Forward modeling:** Built a **Python**-based GIWAXS framework to extract site occupancies, anisotropic Debye-Waller factors, mosaicity, and geometric parameters (APS Mar 2023).
- **Defect analysis:** Extended GIWAXS methods to **model diffuse scattering** from stacking faults and **quantify defect densities** (APS Mar 2024).
- **Film growth:** Grew phase-controlled PbI_2 films via **Chemical Vapor Deposition** while validating polytype fractions (ACS AMI 2023).
- **Scientific ML and AI workflows:** Develop **PyTorch** CNN workflows trained on simulated GIWAXS detector images for structure-analysis screening. The use LLM-assisted agents and reusable AI skills for code scaffolding, parameter exploration, documentation, reproducibility checks, and validation planning against scattering constraints.

Previous Roles

2019–2020 — **Research Intern**, NASA Space Consortium
Synthesized graphene films via PLD/PVD; characterized them with Raman spectroscopy and electron microscopy.

2017–2020 — **Research Assistant**, Missouri State University
Advisors: Dr. K. Ghosh and Dr. S. Mitra
Designed and built a PLD system; characterized thin films by XRD, Raman, SEM/EDS, and profilometry (MRS Spring 2019).

Previous Roles (continued)

2019 — **R&D Intern**, Dynatek Labs
Developed software for biomedical testing and automated hardware systems.

Publications

Planned 2026 David Beckwitt et al., *Quantitative X-ray Diffraction Analysis of 2D-Oriented Powders with Area Detectors*.

Selected Presentations

2023–2024 Invited talks and presentations on X-ray diffraction at APS Prairie Section 2023, APS March Meeting 2024, ACNS 2024, and Missouri State University Seminar 2024.

2020 Fabrication and Characterization of 2D Heterostructures of Graphene and Transition-Metal Oxides.

2017–2018 Undergraduate poster presentations on thin-film materials at Einhellig Forum 2017, Einhellig Forum 2018, and Arkansas INBRE 2018.

Technical Skills

- Core: **Python** (7+ yrs), C++, Fortran, Git, **MPI**, SQL, LaTeX, Excel/VBA
- Data & ML: **NumPy**, pandas, **SciPy**, **PyTorch**, TensorFlow, CNN workflows, scientific image analysis
- Instrumentation: **X-ray / neutron scattering**, **CVD**, PLD, SEM, Raman spectroscopy
- Visualization: **Matplotlib**, Plotly, OriginLab, MATLAB, Jupyter, Dash
- AI-assisted research: LLM-assisted agents, reusable AI skills, code scaffolding, literature triage, documentation, reproducibility checks, validation planning; GitHub Copilot, **OpenAI tools**, Claude, **Perplexity**
- Communication: **Technical writing**, grant proposals, peer review, presentations, video editing

Applied Research Tools

2021–Present

2D Powder Simulator

- Simulates detector-space diffraction from 2D-oriented powder films with geometry and model parameters tied to area-detector images.

2D Powder Visualizer

- Visualizes how orientation distributions and mosaic spread reshape two-dimensional diffraction patterns.

Detector Data Pipeline

- Reads proprietary OSC detector images and converts them into accessible formats for inspection and downstream analysis.

Teaching Experience

2021–Present — **Academic Tutor**, Physics Courses, University of Missouri, Columbia, MO.
Individual and small-group tutoring in undergraduate physics.

2018–2023 — **Teaching Assistant**, University of Missouri and Missouri State University STUDENT REVIEWS.
Taught and assisted physics courses with enrollments from **15 to 200+ students**.
Selected courses: introductory algebra- and calculus-based mechanics; electricity and magnetism; classical mechanics.

2018–2021 — **ACT Prep Tutor**, Club Z!, Springfield, MO.
Individualized test-preparation instruction.

2014–2020 — **Martial Arts Coach**, Dunham's Martial Arts, Springfield, MO.
Led skills instruction and student development across ages and experience levels.

Leadership, Service & Outreach

Affiliation	Service	Leadership	Professional Service	Public Engagement
University of Missouri	• CGW Diversity Officer • Graduate Professional Council	• PAGSA President, 2022–2023 • PAGSA Vice President, 2023–2024 • Director, *PhysAssist*	• Sigma Pi Sigma Physics Congress Judge • X-ray and machine-learning workshops, 2022–present	• MU Extension eclipse events • STEM Cubs • Moberly Correctional Center • Columbia Young Scientist Expo
Missouri State University		• CNAS Leadership Board • Sigma Pi Sigma President		• SPS high school engagement • Physics demonstrations • Student mentoring

Awards

Research	Scholarships / Fellowships	Teaching	Leadership	Travel
• Research & Creative Activity Forum Award, 2025 • NSSA Outstanding Student Research Prize, 2023 • ACNS invited talk, 2023	• Fergason Fund, 2021–2023 • Newell S. Gingrich, 2021–2023 • O.M. Stewart, 2021–2023	• Green Chalk Teaching Award, 2022–2023	• GPC Excellence in Student Leadership, 2022–2023	• Ron Boain & Catherine Rangel-Boain Travel Award, 2023